WISCONSIN
UNIVERSITY OF WISCONSIN–MADISON

Syllabus

Course Information

CS 839, Section 004: Advanced Topics in Reinforcement Learning

Credits: 3

Credit Hour Explanation: This class meets for two 75-minute class periods each week over the semester and carries the expectation that students will work on course learning activities (reading, writing, problem sets, studying, etc) for about 3 hours out of classroom for every class period.

Course Description: Reinforcement learning is the branch of machine learning that studies how an agent can learn from taking actions and receiving feedback in an unknown environment. Reinforcement learning algorithms have demonstrated impressive empirical successes ranging from beating a human world champion at the board game Go to allowing robots to learn difficult manipulation and locomotion skills. This course aims to introduce students to reinforcement learning and topics at the forefront of reinforcement learning research. The first half of the course provides an introduction to RL fundamentals such as temporal difference learning, model-based learning, and policy gradients. The second half covers advanced topics in the RL research literature such as hierarchical and multi-agent reinforcement learning. The course will assume familiarity with probability, statistics, and topics covered in an introductory machine learning class. Familiarity with the Python programming language is recommended.

Requisites: None but the course will assume familiarity with probability, statistics, linear algebra, and topics covered in an introductory machine learning class (linear regression, neural networks, etc). Familiarity with the Python programming language is recommended.

Course Designations/Attributes: N/A

Learning Outcomes

After taking this course, students will:

1. Understand and be able to apply fundamental RL concepts and algorithms.
2. Understand advanced topics in RL research such as deep RL, hierarchical RL, and multi-agent RL.
3. Have completed an RL research project including an implementation component and experimental analysis of implementation.

Meeting and Instructor Information

Meeting Time and Location: Morgridge Hall 2538

Instructional Modality: In-person lectures

Instructor Title and Name: Professor Josiah Hanna

Instructor Contact Information: jphanna@cs.wisc.edu

Day/Time/Location of Office Hours: Tuesdays from 11am - 12pm

Teaching Assistant Name(s): N/A

Teaching Assistant Contact Information: N/A

Day/Time/Location of TA Office Hours: N/A

Overview

Materials

Link to Canvas Site: https://canvas.wisc.edu/courses/477251

Other Digital Instructional Tools (if applicable):

Piazza: https://piazza.com/wisc/fall2025/cs839004/home

Gradescope: https://www.gradescope.com/courses/1109753

Required Textbook(s), Software and Other Materials: Reinforcement Learning: An Introduction (2nd edition). Rich Sutton and Andy Barto. MIT Press, 2018. (Available for free online: http://incompleteideas.net/book/RLbook2020.pdf)

Coursework and Grading

Assignments/Exams/Other Graded Work: See grading below

Exam Proctoring (if applicable): N/A

Grading:

The following weights are used:

- **Weekly Readings and Questions:** 10%
- **Paper Presentation:** 10%
- **Class Participation:** 10%
- **Midterm Exam:** 20%
- **Final Programming Project:** 40%
    - Project Proposal: 5%
    - Literature Survey: 10%
    - Final Report and Code: 25%
- **Programming Assignment:** 10%

Schedule

See the course schedule page: https://pages.cs.wisc.edu/~jphanna/teaching/25fall_cs839/

AI Policy

The class policy is generally to allow and encourage use of AI as a tool that can help you go farther with your final projects and for additional study.

The one exception to this permissive policy is the weekly pre-class reading responses. As the pre-class reading is critical to effective class participation, use of AI to bypass the reading and response writing is not allowed. If AI use is suspected (due to the response itself or a lack of preparedness for class period), the response will receive a zero.

Otherwise, you are welcome (but not required) to use AI tools to complete programming assignments, refine writing, etc. However, all submitted content will be treated as your own, including for the purpose of determining academic dishonesty.

Please document your use of AI and include your documentation with your assignment submission. Documentation could include specific prompts used, process for iterating on prompt design, etc.

Academic Policies and Statements

- Academic Calendar and Religious Observances
- Academic Integrity
- Accommodations for Students with Disabilities
- Course Evaluations
- Diversity and Inclusion
- Mental Health and Well-Being
- Privacy of Student Records and the Use of Audio Recorded Lectures
- Remote Proctoring with Honorlock (if applicable)
- Students' Rules, Rights and Responsibilities
- Teaching & Learning Data Transparency